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Overview of Big Data

Defining Big Data:

As per Oxford English Dictionary, the definition of Big Data is "data of a very large size, typically to the extent that its manipulation and management present significant logistic challanges".

In the context of information technology and data science Big Data means technologies and initiatives that involve data that is ~~too diverse~~

i) too diverse

ii) fast changing or massive for conventional technologies and infrastructure to address efficiently.

Data analytics and artificial intelligence along with rapid development of hardware and software technology made Big Data a hottest trend in the industry.

Ex: i) A retailer can track user web click to identify behavioral trends that improve campaigns, pricing and storage.

ii) Utilities can capture household energy usage energy levels to ~~track~~ predict outages and to incentivize more efficient energy consumption.

iii) ~~Sefologist~~ can do sentiment analysis using social media data to predict the result of election.

iv) Government and even Google can detect and track the emergence of disease outbreaks.

are some of the

At this juncture we can define Big Data as follows:

Best Practices Definition:

The emerging technologies and practices that enable the collection, processing, discovery, analysis and storage of large volumes and disparate types of data, quickly and cost effectively.

The important features of Big-Data are expressed as 4V's.

i) Volume ii) Variety iii) Velocity and iv) Value or veracity.

Volume: This signifies large amount of data. Typically when people discuss big data volumes, the discuss petabytes (10^{15} byte). But in reality, most real-life Big Data implementations are still in the 10's to 100's of terabytes (10^{12} bytes). Which is still a lot of data.

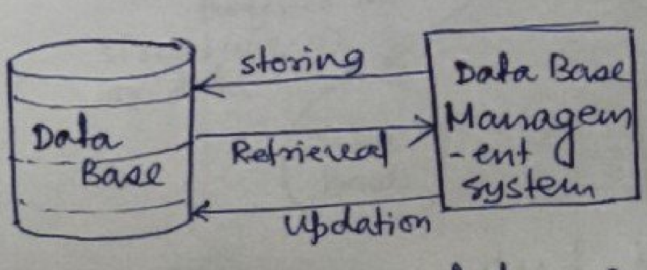
Variety: This refers to the evolving types and growing sources of data. It could be either structured or semi-structured or unstructured data.

Velocity: Velocity in the content of big data refers to the speed of data acquisition and processing. ~~Big data technologies~~ Ex: Every 2 days face book acquire 4 Petabyte of data.

Value or Veracity:
In an effort to distinguish the business value of big data, the vendor community ~~come~~ came up with a ~~4~~ "4th" V — value or veracity.

The fundamental entity in this subject is the data which is the raw information gathered about an object in the real world.

In pre Bigdata era the data were collected in a database and updating the data or accessing the data or their storage was managed by a database management system. which can be represented schematically as—



Now to store the data a number of structure or model are used. Namely—

i) Hiererchical ii) Network model iii) Relational model
iv) Entity-Relationship model.

All these model are developed to establish correlation between data.

In short the data model is an abstract model that orgaizes elements of data in a certain format. It means when the data is stored in a database it maintain certain format.

Therefore data model describes the method or structure used to store the data in the data base.

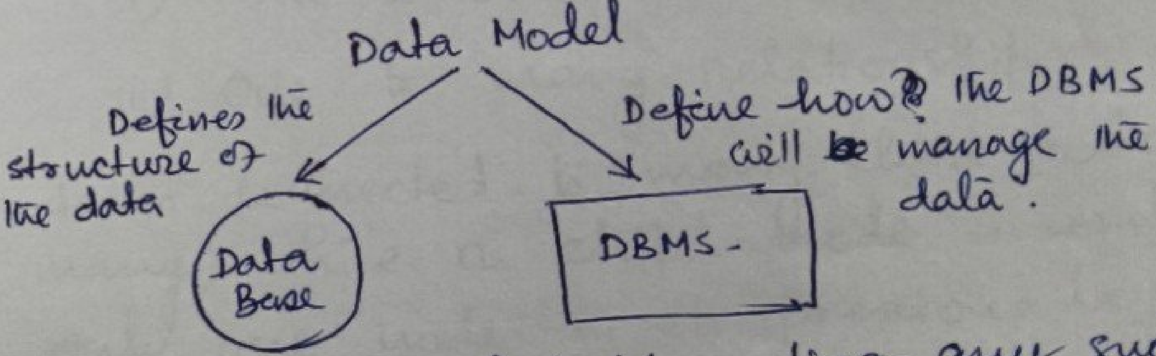
In any model the data in the database are related to one another. For example—

Hospital staff

Data is related to one another

Doctor Id.	First Name	Last Name	Phone No

Data model defines two things

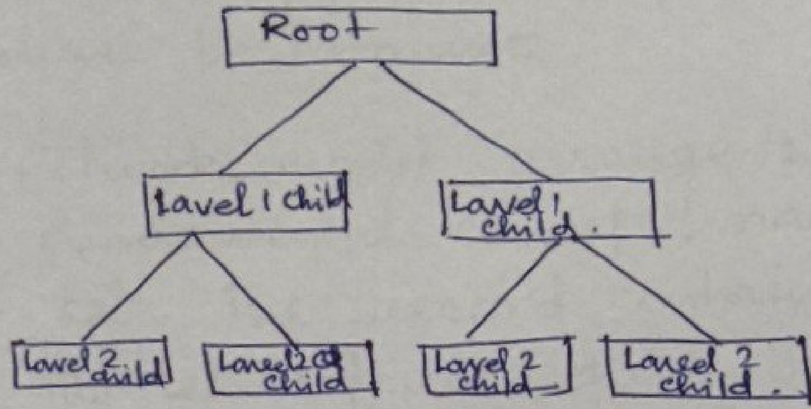


In designing any and implementing any such model there are mainly three stages—

1. Required Attributes
2. Database Designer Design
3. Database Administrator Implement the design.

Hierarchical Model

in hierarchical model the data are arranged in a tree like structure.



Hierarchical model has only one way approach. and to retrieve a data the whole tree had to be traveersed.

in hierarchical model two types of relationship exists —

i) one to one relationship

ii) One to many relationship.

A root node is connected to many child node (one to many) while a child node is connec-ted to only one node in its previous layer. (one to one).

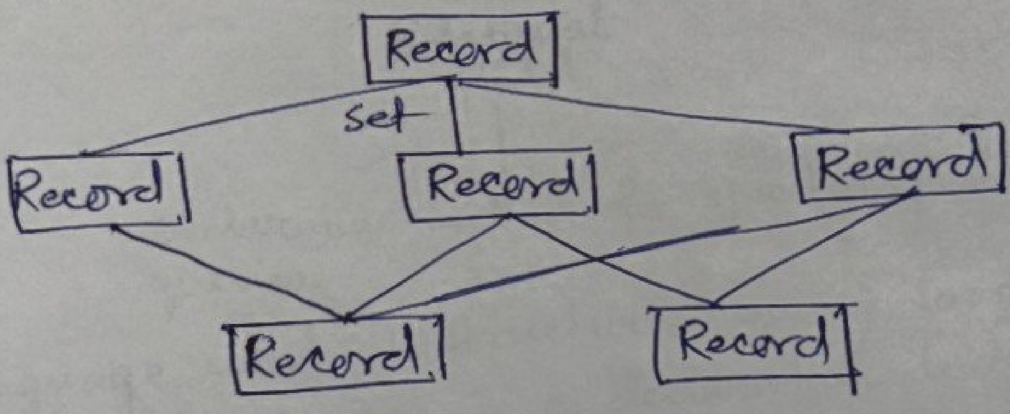
In hierarchical model Many to one or Many to many relationship can not be defined-

This is one of the drawbacks of hiarerchical model. It is useful in nested sort of information. For examp table of contents or ~~Recip~~ recipe chart.

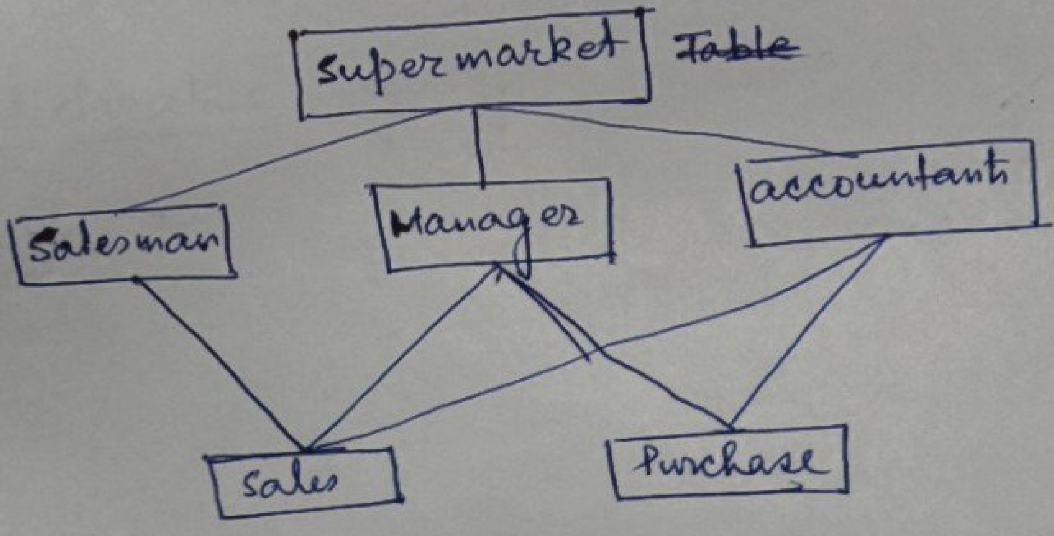
Network Model:

Network model emerged from the hiararchical model. to overcome the disadvantages of the hierarchical model. It represents complex data relationship to improve ~~data performance~~ database performance.

The network model arrange the data using two fundamental concepts, namely, record and set. The record contains the fields and the set defines one to one, one to many, many to one and ~~mem~~ many to many relationship among the fields



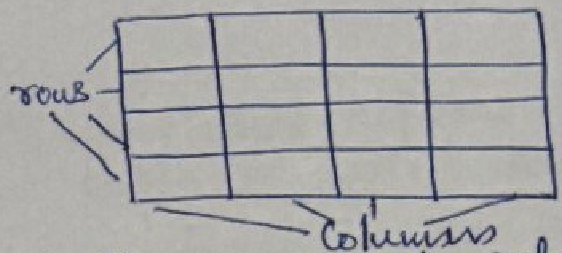
Ex:-



Relational Model

In relational model, data are stored in the form of a two-dimensional table.

Table



Inside the table rows and columns are interrelated. This type of structure gives the most flexible method to ~~updat~~ store, retrieve and update data.

Relational model is implemented through relational database management system or RDBMS.

Relational Model

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Declarative method for specifying data and queries.

The language used is structured query language or SQL in short. **SQL** is a declarative language.

<u>what is declarative language?</u>

Relational diagram: A relational diagram is a representation of the main components of the relational data base.

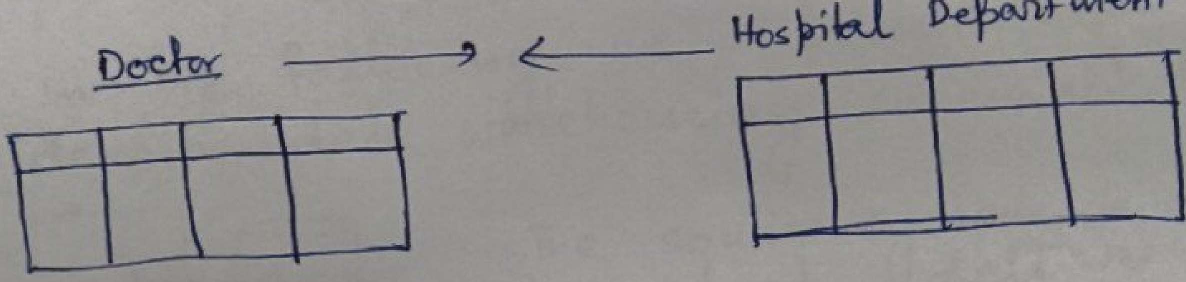
Let's what are the main components of a relational data base.

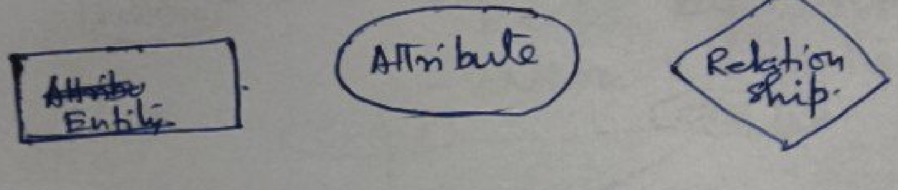
The two dimensional representation of data in the form of a table is called relation. ~~in In~~ Entity defines any object in the ~~Rel~~ real world ~~ab~~ about which the table is created.

Each column of a table describe a particular attribute of an entity and in a row all the attributes about a member of the entity set is stored.

In each column we store same type of data
for ex: in a column of name we can not store any number.

To establish ~~different~~ relationship between diffent entities we use entity-relationship diagram.





All we have stated so~~fa~~ far is related to structure ~~of~~ through which data is stored in a data base, now we will discuss computational infrastructure to implement it.

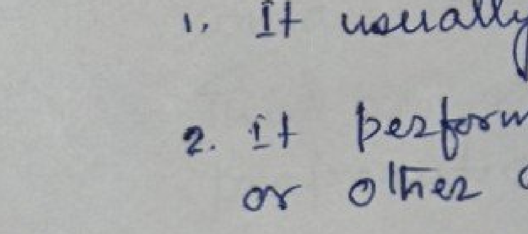
1. Datā Warehouse

A datā ~~ware~~ warehouse is a system that retrieves and consolidates datā periodically from the source systems into a dimensional or normalized datā store.

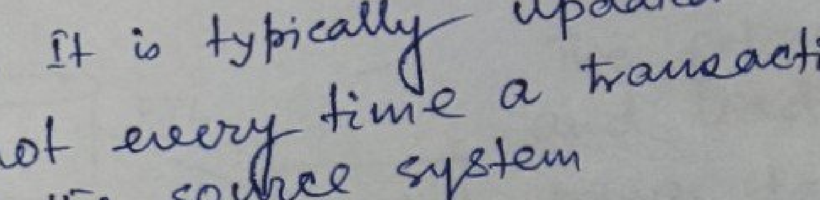
Tasks executes by datā ware house

1. It usually keeps years of history
2. it performs queries, business intelligence or other analytical activities.
3. It is typically updated in batches and not every time a transaction happens in the source system — OLAP.

In the following we represent and describe datā warehousing schematically.



The source system is the individual datā records keeps by different organization in the network.



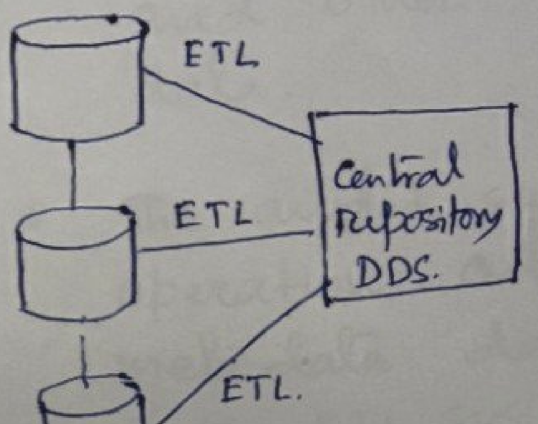
The users interacts with the each of the source system through OLTP (Online Transaction processing)

The mainpurpose is to capture and store the business transactions on real time.

The source systems' data is examined using a data profiler.

Data profiler is a tool that has the capability to analyze data, such as how many rows are in each table, how many rows contain ~~null~~ Null values and so, on.

This process can be termed as data preperation and data cleaning.



The extract transform and load (ETL) then brings data from various source systems into a staging area. The staging system is a dimensional data store (DDS).

A DDS is a database that stores the data in a different format than OLTP. The dimensional format is more suitable for analysis

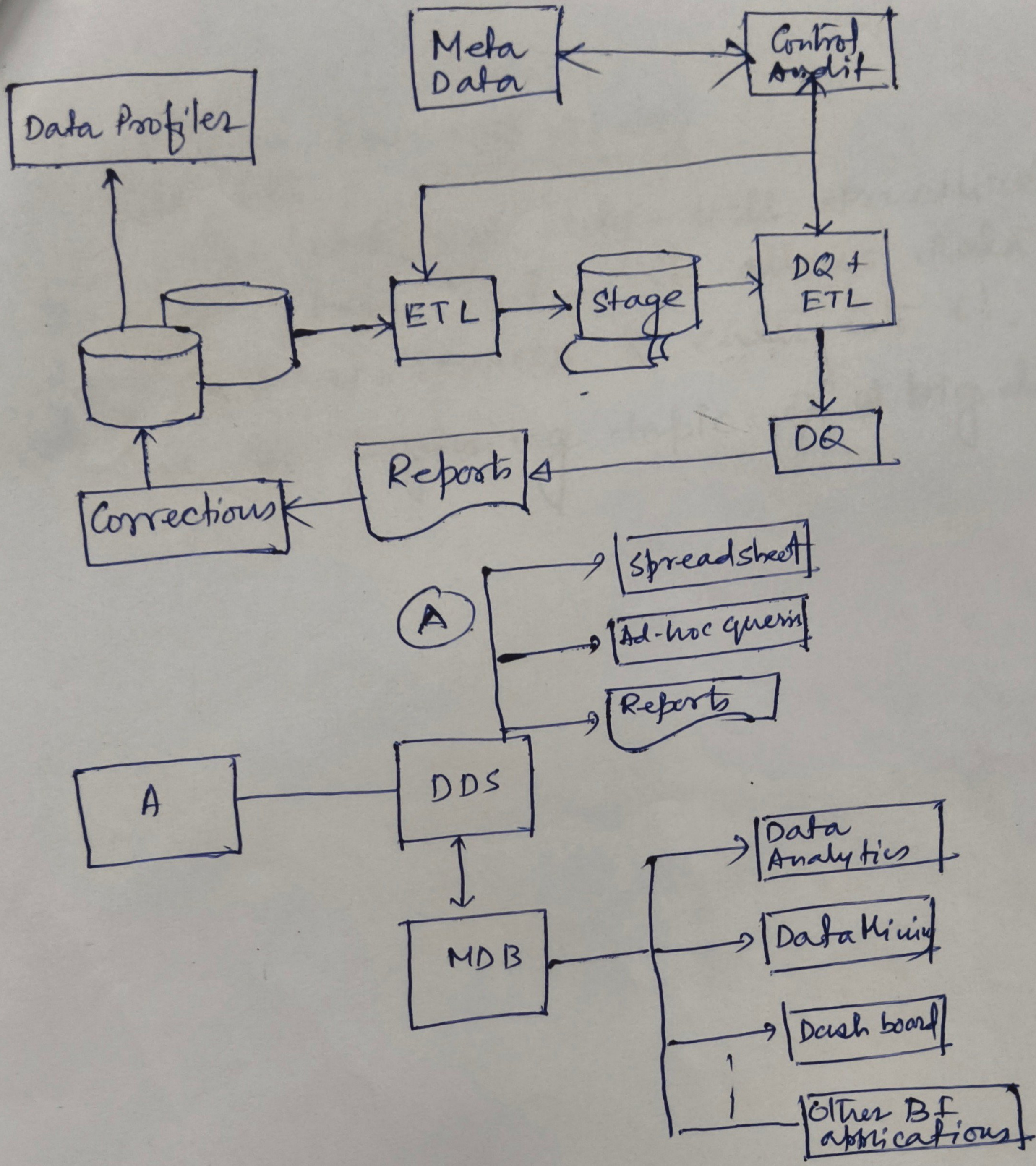
* When the ETL system loads the data into the DDS the data quality rules do various data quality checks.

*) Bad data is put into the data quality database (DQ) to be reported and corrected in the source system.

* The ETL system is managed and or-chestrated by the control system based on the sequence, rules and logic stored in the metadata.

* The metadata is a database contai-ning information about the data structure, the data meaning, the data usage, the data quality rules and other information about the data.

* The audit systems logs the system operations and usage into the metadata database. It is a part of the ETL system that monitors the operational activities of the ETL processes and logs their operational statistics.



This is called the data ware housing system.

Data Mining

Data mining is the process of exploring data to find the pattern and relationships that describe the data to predict the unknowns or future values of the data.

Data lakes: (Microsoft Azure)

in data lake not only well structured data are kept. But it allows data from different sources to enter into it.

This is the begining topic of big data.